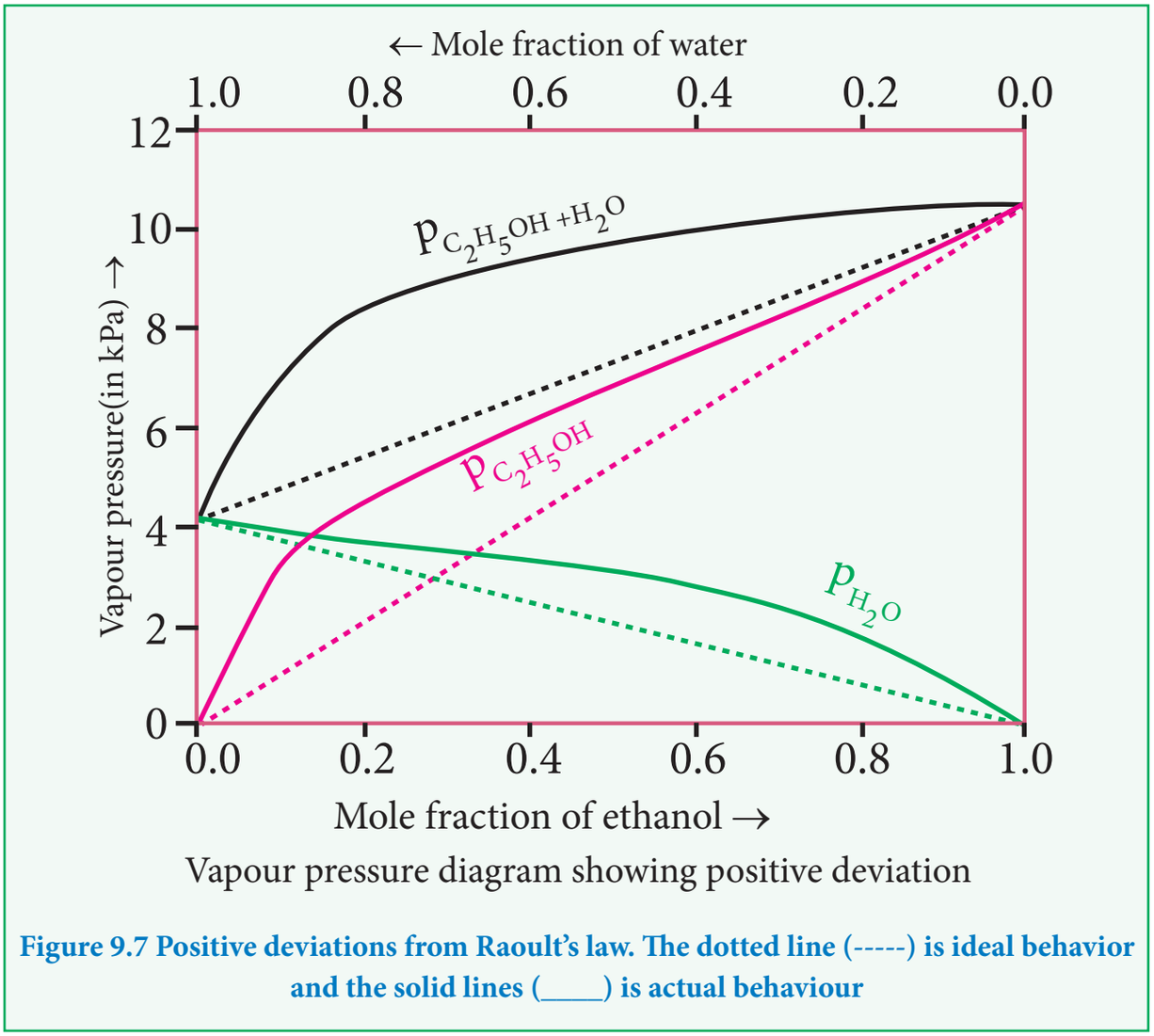

Figure 9.7 Positive deviations from Raoult's law. The dotted line (-----) is ideal behavior and the solid lines (____) is actual behaviour

Non-ideal solutions - negative deviation from Rauolt's Law:

Let us consider a case where the attractive forces between solute (A) and solvent (B) are stronger than the intermolecular attractive forces between the individual components (A-A & B-B). Here, the escaping tendency of A and B will be lower when compared with an ideal solution formed by A and B. Hence, the vapour pressure of such solutions will be lower than the sum of the vapour pressure of A and B. This type of deviation is called negative deviation. For the negative deviation $p_A < p^\circ_A x_A$ and $p_B < p^\circ_B x_B$.

Let us consider a solution of phenol and aniline. Both phenol and aniline form hydrogen bonding interactions amongst themselves. However, when mixed with aniline, the phenol molecule forms hydrogen bonding interactions with aniline, which are stronger than the hydrogen bonds formed amongst themselves. Formation of new hydrogen bonds considerably reduce the escaping tendency of phenol and aniline from the solution. As a result, the vapour pressure of the solution is less and there is a slight decrease in volume ($\Delta V_{mixing} < 0$) on mixing. During this process evolution of heat takes place i.e. $\Delta H_{mixing} < 0$ (exothermic)

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